WASHIM AKRAM

SOFTWARE ENGINEER | BACKEND & DATA INTEGRATION

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Portfolio: https://akramswe.github.io
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PORTFOLIO GITHUB LINKEDIN

PROFESSIONAL SUMMARY

Software Engineer with 3+ years of professional experience across backend applications, SQL databases, ETL workflows, enterprise integrations and financial/payment systems. Experienced with Java, Spring Boot, Microsoft SQL Server, stored procedures, data processing, REST/SOAP APIs, production troubleshooting and release validation. Developed independent microservices using PostgreSQL/MongoDB, automated testing, Docker and GitLab CI/CD. Completing an M.Sc. in Computer Science at the University of Vienna; seeking full-time opportunities in software engineering, backend systems and data integration.

TECHNICAL SKILLS

Programming & fundamentals: Java 17, SQL, Python, JavaScript, C/C++; OOP, algorithms, software design principles

Backend & application development: Spring Boot, J2EE, REST/SOAP APIs, JDBC, Maven, microservices

Databases, SQL & ETL: MS SQL Server, PostgreSQL, MySQL, MongoDB, stored procedures, ETL workflows, data processing, Spring Data JPA / MongoDB

Data & system integration: REST/SOAP, JSON, XML/XSD, XSLT, XPath, XQuery, data transformation

Reactive & concurrent Java: Spring WebFlux, Project Reactor (Mono/Flux), CompletableFuture, ExecutorService, `@Async`

Testing & delivery: JUnit, Mockito, unit/integration/API/system testing, Docker, Docker Compose, GitLab CI/CD, Git/GitHub, Postman

Security & practices: HashiCorp Vault, Linux, Windows, debugging, distributed systems, Agile / Scrum

PROFESSIONAL EXPERIENCE

Software Engineer | Limerick Resources Ltd. Jan 2023 – Dec 2023
Dhaka, Bangladesh (Hybrid)

- Developed and maintained Java/Spring Boot functionality across enterprise application services, backend and frontend components, and database-backed workflows.
- Implemented Spring Boot REST controllers and service-layer components using dependency injection and application configuration.
- Maintained J2EE components and SOAP-based enterprise integrations using XML/XSD message formats.
- Used JDBC for application data access; diagnosed connectivity issues between application and database layers.
- Developed enterprise data integration and transformation functionality using JSON, XML/XSD, XSLT, XPath and XQuery.
- Resolved backend, database and integration issues; tested and validated changes before delivery.
- Collaborated with 5+ engineers on development, testing and delivery, contributing to approximately 25% faster delivery.

Software Engineer | Nassprotech Mar 2022 – Dec 2022
United States (Remote)

- Supported payment and financial applications across Java services, Microsoft SQL Server and production data workflows.
- Maintained SQL services, stored procedures, ETL workflows and payment-related data processing.
- Worked with HashiCorp Vault for application credentials and environment-specific production configuration.
- Investigated production incidents across application services, databases and payment-processing workflows alongside international teams.
- Analysed payment and transaction data issues and validated changes after application and database fixes.
- Supported release testing and verification of application behaviour, data processing and production stability.

Junior Software Engineer | Nassprotech Aug 2020 – Mar 2022
United States (Remote)

- Supported production Java financial and payment applications, including backend data services.
- Worked with MS SQL Server, queries, stored procedures, ETL processes, reporting systems and financial transaction data.
- Investigated payment application, database and operational issues with senior engineers; supported testing and maintenance after updates.

PROFESSIONAL EXPERIENCE (CONTINUED)

Lab Teaching Assistant | Daffodil International University Jan 2019 – Apr 2020
Dhaka, Bangladesh (On-site)

- Supported Java, object-oriented programming and software development lab sessions.
- Helped students debug Java code and understand classes, inheritance, interfaces and exception handling.
- Reviewed programming assignments and technical reports and provided implementation feedback.

SELECTED SOFTWARE ENGINEERING PROJECTS

Payment Processing Microservice

Java 17 | Spring Boot | WebFlux | PostgreSQL | https://github.com/akramswe/payment-processing-microservice

- Developed payment creation APIs with request validation, status transitions, transaction history and error handling.
- Implemented reactive REST endpoints and asynchronous workflows using Spring WebFlux, Project Reactor (Mono/Flux) and Java concurrency APIs.
- Used PostgreSQL and Spring Data JPA for persistence while applying reactive processing at API and orchestration layers.
- Added idempotency checks, exception handling and structured logging for payment workflows.
- Integrated HashiCorp Vault for secrets and environment-based configuration.
- Wrote unit/integration tests using JUnit and Mockito; automated builds and tests with Maven, Docker Compose and GitLab CI/CD.

Patient Management Microservice

Java 17 | Spring Boot | MongoDB | https://github.com/akramswe/patient-microservice-spring-boot

- Built REST CRUD APIs with controller, service, repository and persistence layers using Spring Data MongoDB.
- Added JUnit/Mockito tests and configured Maven, Docker, Docker Compose and GitLab CI/CD workflows.

ACADEMIC RESEARCH

University of Vienna | LLM-Assisted Program Analysis

- **Algorithm identification in obfuscated C programs:** Developed a Python pipeline combining static semantic features, rule-based classification and LLM-assisted analysis; checked predictions against JSON ground truth after Tigress control-flow obfuscation.
- **Reconstructing long function-call chains:** Built a Python framework to generate C programs and evaluate LLM reconstruction of call chains up to 1,000 functions, including indirect calls, function pointers and switch-based dispatch.

EDUCATION

M.Sc. Computer Science | University of Vienna, Austria (2024 – Present, final semester)

Thesis: Trust-Boundary Enforcement for Agentic Reverse Engineering of Obfuscated Binaries

B.Sc. Software Engineering | Daffodil International University, Bangladesh (2016 – 2020)

CGPA: 3.95 / 4.00 | German equivalent: 1.1

LANGUAGES

English – C1 | German – B1 (working toward B2) | Bengali – Native